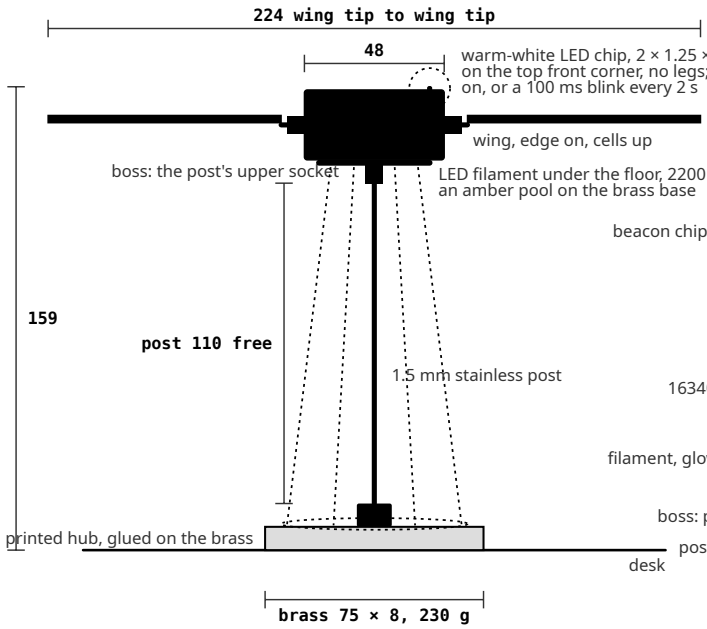
Desk satellite: how the finished thing looks

A black bus on one stainless post over a brass base, two solar wings, a tiny warm light on the top front corner, an amber pool below. 224 wide × 45 deep × 159 mm tall, about 328 g. Dimensions in mm; each view at its own scale.

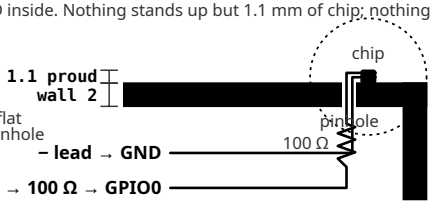
Front view (lid and USB port on the far side)



The beacon at the corner, 8 px per mm

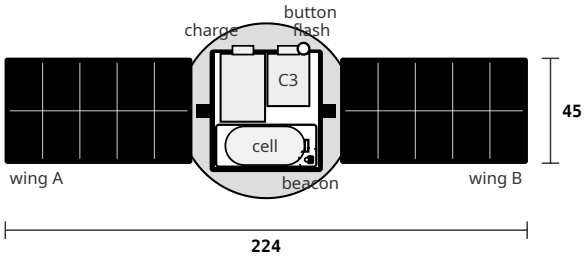
section through the top front-right corner, seen from the front

0805 chip, 2 × 1.25 × 1.1 mm, glued flat on the top 5 mm in from the front and the right face;
its own 0.1 mm enamelled leads run 1 mm over the top and down a 1.2 mm pinhole to the resistor
and GND inside. Nothing stands up but 1.1 mm of chip; nothing is soldered to the box or to steel.



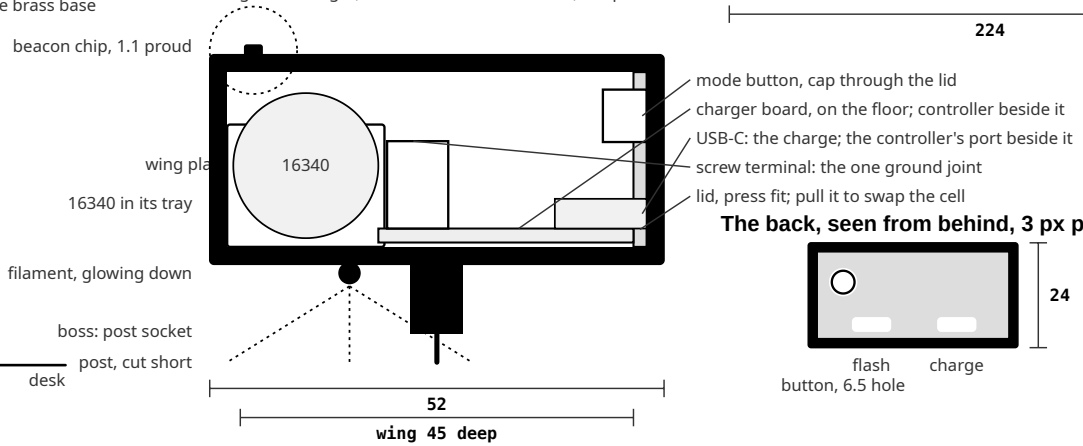
Top view

front of the desk at the bottom; the brass shows around the bus

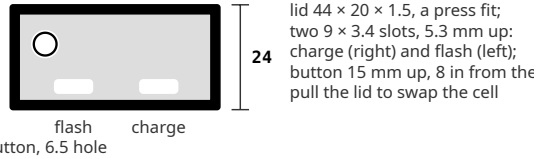


Side section through the bus, front on the left

cut through the charger; the back is the service side; the post cut short



The back, seen from behind, 3 px per mm



Key numbers

224 × 45 × 159 mm	overall, wings to the top of the LED
48 × 52 × 24 mm	the bus: printed black, 2 mm walls
2 × 80 × 45 wings, 5 V, series	12.2 V in sun, 7 to 8 V in a room
328 g, 230 in the base	centre of mass 46 mm up; tips at 39°
3 rods, 3 CA sockets	post and two booms; no frame joints
7 wires, 2 resistors, 1 diode	30 AWG; wings and beacon have their own leads
0.22 / 0.42 / 3.4 mA	asleep / beacon / glow, from the cell
84 / 10 / 2.6 days	beacon / glow / lamp, no light at all
USB-C 500 mA, 2.2 h	the fallback charge, through the lid

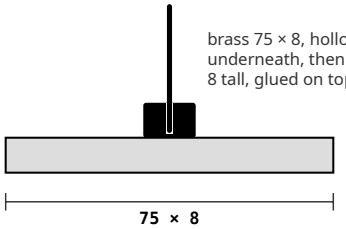
50 mm in the front view

What the wings deliver (listing figures, scaled by light)

where	lux	pair, mA	stored a day
desk, room light only	500	0.3	0 mAh
beside a window, dull day	1,000	0.9	1 mAh
beside a window, bright day	3,000	2.7	12 mAh
on the sill, overcast sky	10,000	9.0	50 mAh
an hour of direct sun	100,000	90.0	89 mAh / h

The charger spends 0.75 mA of input on itself and gives 6 h a day:
'stored' is six hours of what is left. Room light stores nothing.
Beacon mode needs 10 mAh a day, glow 82: a bright window carries the beacon,
the amber pool lives on direct sun or the cable; a USB top-up now and then.

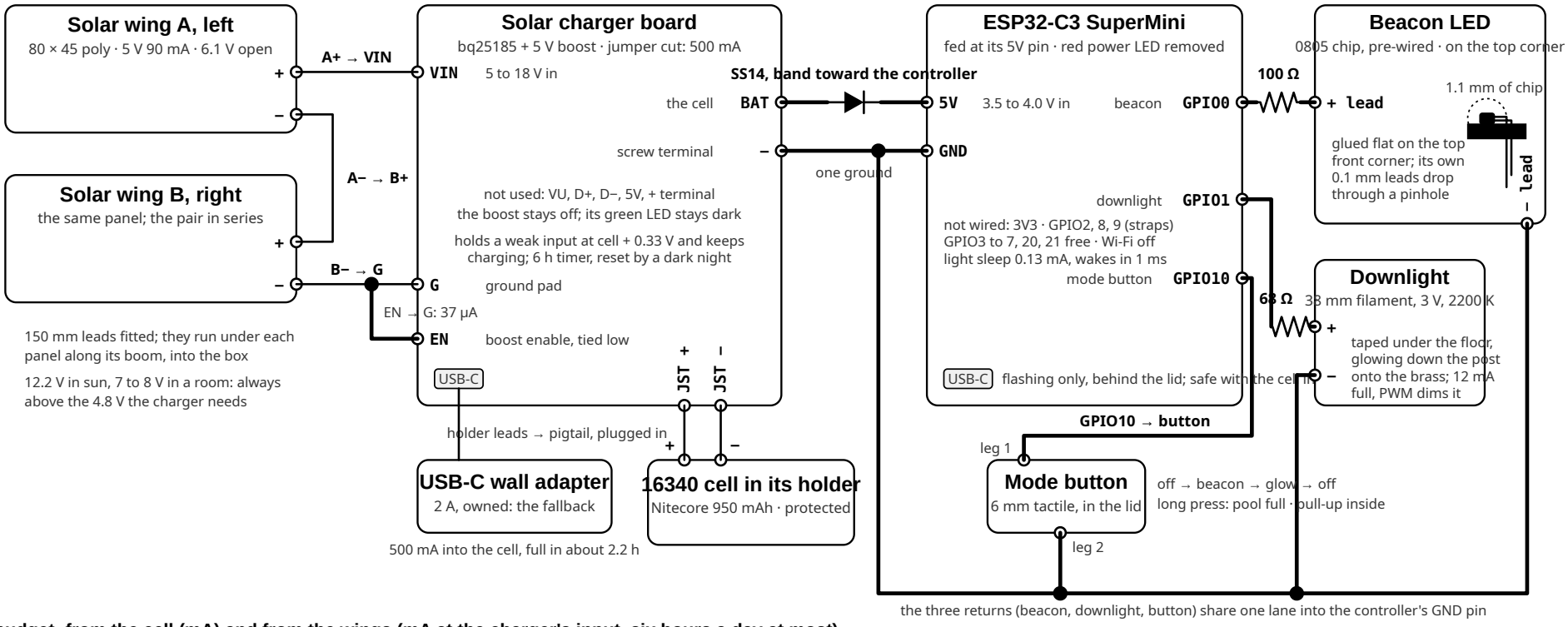
The base, 3 px per mm



Paper design 2026-11-02, revised twice that morning, approved 2026-11-03 and revised for the 16340 cell. The box is an envelope round the real holder and boards: bench-check before printing. Nothing built.

Desk satellite: every connection on one sheet

A dot is a joint. A line that hops over another does not touch it. Thick = 30 AWG wire you solder. Thin = a lead or cable that comes with the part (the wings' and the beacon's leads are thin lines here). Pins are grouped by job, not by position o



The budget: from the cell (mA) and from the wings (mA at the charger's input, six hours a day at most)

asleep	`0.22 mA`	`5 mAh/day`	controller in light sleep, regulator, the EN pull-up, the charger's 4 µA
beacon	`0.42 mA`	`10 mAh/day`	the chip, 4 mA for 100 ms every 2 s · 84 days on the cell in the dark · the resting mode
glow	`3.42 mA`	`82 mAh/day`	beacon blinking + the pool dimmed to 3 mA · 10 days in the dark · evenings and sunny sills
lamp	`13.42 mA`	`322 mAh/day`	pool flat out at 12 mA, beacon steady · 2.6 days · a long press; plug in for it

Wings: 0.3 mA under room light, nothing stored (the charger eats 0.75 mA itself) · 0.9 to 2.7 mA beside a window, 1 to 12 mAh a day · 9 mA on an overcast sill, 50 mAh a day · 90 mA in full sun, the listing's figure, scaled by lux; the first week on the sill, with the meter on the BAT pad each evening, is the measurement.

Mods, both before wiring: cut the charger's charge-rate jumper (1 A → 500 mA, half the cell's rating an hour); lift the SuperMini's red power LED, or its resistor, with tweezers (about 1 mA all day: beacon mode would last 25 days, not 84). Never: the cell on anything but the charger's JST socket · a USB cable in the controller without the diode in place · the wings on the charger's USB side · the filament without its resistor · an iron within 10 cm of the cell.

Paper design 2026-11-02, not built. 0 crossing(s) on this sheet. Sources: TI bq25185 datasheet SLUSF65B, the Adafruit product page, guide, schematic and fab print, the SuperMini schematic notes, the three listings.

DESK SATELLITE · STEP 1: THE FINAL BUILD, APPROVED 2026-11-03

A satellite on a post, blinking warm

How it looks, how it is wired, every part and every pin. 224 × 45 × 159 mm, about 328 g, no solder in the frame or on the wings. Paper design of 2026-11-02, revised the same morning at the builder's word: the beacon is a tiny LED chip glued on the corner of the bus, no legs, no mast; the hologram's amber filament glows down from under the body. Approved 2026-11-03 as drawn, then revised the same day at the builder's word for a Nitecore 16340 cell in its holder in place of the pouch (the bus grew round it); the science and engineering behind it is Desk Satellite Physics. Nothing printed or built.

What it is

A small satellite on a post. Two solar panels, 80 × 45 mm, are its wings, a matte black printed box is its body, and a tiny warm-white LED chip is glued on its top front corner, no legs showing: on, or blinking once every two seconds, like a satellite you can see pass over. Under the body the hologram's amber LED filament glows down the post onto a brass base, the family's metal. Inside the box the owned charger board tops up a Nitecore 16340 cell in its holder from the wings, with a USB-C port through the back as the fallback, and an ESP32-C3 SuperMini sleeps between blinks. One 1.5 mm stainless post carries it over a solid brass base, 75 mm across.

The reference is Huy Vector's Desk Satellite: a copper-wire cube with an ESP32-C3 clock and a 0.96-inch screen inside, two solar wings, a dish, a charging module and a separate solar controller, on a brass tube over a bamboo coaster. Ours keeps the silhouette and the solar wings and drops the clock: the builder's brief is panels, a cell they top up, USB as the fallback, and a small 3000 K light at the tip that is on or blinks.

This page is step 1 of the house rule: the final build first, approved by the builder on 2026-11-03 as drawn, and revised the same day at their word for the Nitecore 16340 cell they had bought, in place of the 500 mAh pouch. One form sheet, one wiring sheet, every part and every pin. The science behind it (what a solar cell does in room light, how the charger follows a weak wing, the beacon's eye, the post as a column) is step 2, Desk Satellite Physics. Every part is bought or in a drawer; nothing printed, cut or soldered yet.

The decisions

A SATELLITE AT A GLANCE: BODY, TWO WINGS, A LIGHT ON THE CORNER

The reference is a copper-wire cube with a clock inside, two solar wings and a dish, on a brass tube over a coaster. Ours keeps that silhouette and nothing it does not need: a matte black bus 48 × 52 × 24 mm, two 80 × 45 mm panels on short booms, a tiny warm light glued on its top front corner, an amber pool under the body on a brass base, one stainless post, one brass base. 224 mm wing tip to wing tip, 159 mm tall, about 328 g. No display, no clock, no sensor: the builder's brief is panels, a cell they keep topped up, USB as the fallback, and a 3000 K beacon at the tip that is on or blinks.

TWO 5 V WINGS IN SERIES, IN A RECTANGLE

The charger holds a weak input at the cell's voltage plus 0.33 V and will not charge below that, and the board adds a Schottky, so the wings must deliver 4.3 to 4.7 V in whatever light they get. One 10-cell 5 V panel opens at 6.1 V in sun and about 4 V in a bright room: too low exactly when it matters. Two in series open at 12.2 V in sun and 7 to 8 V in a room, always above the line, and the charger pulls the pair down and takes their full current, 90 mA in sun; the board is rated to 18 V. The wings changed at the builder's word on the evening of 2026-11-03: the sheet was approved with two 56 mm squares; the builder then wanted a rectangle a little smaller than the reference's 110 × 60 mm panel, and Amazon's one such panel with leads is the 80 × 45 mm 5 V 90 mA poly with 150 mm leads welded on: the same current as the squares, nothing to solder on the panel, 224 mm wing tip to wing tip.

CYBER DAWN'S SPARE CHARGER BOARD, STRAIGHT FROM THE DRAWER

Cyber Dawn bought two bq25185 boards and dropped solar on 2026-10-31 because a 5 W panel could not feed a clock whose display alone drinks 50 mA (its `SOLAR_FEASIBILITY.md` is the record). Here the whole satellite sleeps at 0.2 mA and blinks at 0.4, a hundred times less, so the same board and a pair of 80 × 45 mm panels close the budget at a bright window. What the board gives, from its schematic and TI's datasheet: USB-C and a 5 to 18 V VIN pad through two Schottkys, so the panels and a cable never fight; charging that follows a weak input instead of collapsing it; no false

termination while it does so; a 6 h timer that a night resets; 4 µA from the cell when idle; 500 mA with the jumper cut. Its 5 V boost converter is switched off with one wire (EN to G) and its screw terminal becomes a ground point.

A 16340 IN A HOLDER, NEVER SOLDERED

The sheet was approved with the owned 500 mAh pouch. The same day the builder chose a Nitecore 16340 cell instead, thirteen in the drawer from two IMR Batteries orders of 2026-10-16: 950 mAh, protection under the wrap, button top, 20 g. 84 days of blinking in the dark instead of 45, 10 of glow instead of 5, and a cell that lifts out of its holder when the lid is off. The holder is Cyber Dawn's 43 × 18 × 14 mm tray; its leads meet an owned JST-PH pigtail 40 mm from the cell, so no iron comes near it, and the plug goes into the charger's socket. The price is the box: the bus grows from 42 × 41 × 22 to 48 × 52 × 24 mm to take the tray along its front wall with the two boards behind it, both ports through the lid, and the satellite gains about 25 g.

THE CONTROLLER RUNS FROM THE CELL THROUGH ONE DIODE, AND THE BOOST STAYS OFF

The board's BAT pad is the cell. One SS14 from it to the controller's 5V pin gives the controller 3.5 to 4.0 V, which its own regulator turns into 3.3 V, and costs 0.15 V. The diode is not for the charger: on the SuperMini the 5V pin is USB VBUS with no diode, so without it plugging a flashing cable would push 5 V straight into a lithium cell. With it, flash whenever you like. The charger's 5 V boost would be the tidy way to power a 5 V board, but it idles at about a milliamp with its green LED, five times the sleeping controller; one wire from EN to G turns it off for 37 µA through its pull-up.

THE ESP32-C3 SUPERMINI AGAIN, WITH ITS POWER LED GONE

Thirteen are owned, the toolchain exists, and the firmware is a hundred lines: light sleep with a timer, a pin, a button. The chip sleeps at 0.13 mA in light sleep and wakes in a millisecond, which is what a blink wants; deep sleep would spend more rebooting every two seconds than it saves. The board's red power LED burns about 1 mA all day, five times the rest of the sleeping satellite, so it comes off with tweezers before anything is wired. An ATtiny would sleep at 1 µA and make a desk under room light nearly self-sustaining, at the price of a new toolchain and programmer; written down here as the upgrade if the window budget ever feels thin.

THE BEACON: A CHIP GLUED ON THE CORNER, NO LEGS, NO MAST

The first draft put a 3 mm LED on two bare rods 52 mm above the bus. The builder wanted a small light stuck on the corner, not a tall pin, with no legs at all, so the mast went the same morning and the LED became a pre-wired 0805 chip: 2 × 1.25 × 1.1 mm, with two 0.1 mm enamelled leads already soldered on. It is glued flat on the top of the bus 5 mm in from the front and the right wall, its leads run 1 mm over the surface and down a 1.2 mm pinhole, and the resistor sits inside above the controller: nothing stands up but 1.1 mm of chip, and nothing is soldered to the box. A warm-white chip drops about 2.9 V, so 100 Ω from the 3.3 V pin gives 4 to 5 mA, a clear point of light in a room (the kit's 68 Ω gives 7 mA if it looks faint; the chip takes 20); PWM dims the steady mode to about a quarter. The overall height falls from 209 to 152 mm.

AN AMBER POOL UNDER THE BODY

The builder asked for a light below the satellite that shines down, in the warm yellow that matches the family's brass. The hologram's LED filament, the 38 mm 2200 K strip with nine spare in its pack, is Kapton-taped under the floor of the bus 10 mm ahead of the post, parallel to the wings, glowing down the post onto the brass base 118 mm below: a pool of amber on brass. GPIO1 drives it through 68 Ω, about 12 mA flat out (the hologram's figure), PWM-dimmed to 3 for the glow mode; its two leads come up through a 1.5 mm hole in the floor. It is the one thing on this satellite that is not for the satellite's own sake, and it costs: the glow mode is 3.4 mA, 10 days on the cell in the dark, so it is the window and evening mode, and the beacon alone is the one that lives on the wings.

ONE POST ON A WEIGHTED BRASS BASE, A PRINTED HUB AND BOX: NO SOLDER IN THE FRAME

A brass coaster, 75 × 8 mm, is the base. It weighs only 80 g, a thin top over a hollow, so about 150 g of steel is epoxied into the hollow and the pad goes back over it: 230 g, all of it in the bottom 8 mm, unseen. A 12 mm printed hub glued to its flat top is the post's socket, since the top is too thin to drill. It replaced the 3 in stainless disc on 2026-11-06 at the builder's word: the steel read too white on the bench, and brass is the family's metal and the amber pool's colour. The post is one 1.5 mm coupon, 110 mm free, into a boss under the box. Each boom is a 1.5 mm rod in a 6 mm boss on the box's side wall, glued along the back of its panel. Three sockets, three drops of CA, zero frame joints, no solder on steel anywhere. Centre of mass 46 mm up on a 75 mm base: it tips at about 39°, against 30° on the steel disc and about 25° on the coaster

unweighted. The box is the one printed body in the family after Cyber Dawn's, and for the same reason: a satellite bus is a box.

WHAT THE PANELS REALLY DO

The charger itself takes about 0.75 mA from its input while charging, so below about 1 mA of panel current nothing is stored. Room light on a desk (500 lux) gives the pair about 0.3 mA: nothing. Daylight beside a window (1 000 to 3 000 lux) gives 0.9 to 2.7 mA, so 0.2 to 2.0 mA net, which over the charger's 6 h window is 1 to 12 mAh a day against the beacon mode's 10: a bright window carries it, a dull one does not. An overcast sill (9 mA, 50 mAh a day) banks five beacon days in one; the glow mode's 82 wants hours of direct sun or the cable. On a desk away from the window it is a 84-day battery object that gets a USB top-up; at a bright window it is a solar one. These are the listing's full-sun figure scaled by lux (its 38 000 lux was a halogen lamp, which silicon counts at nearly three daylight lux each), not measurements; the first week on the sill, with the cell voltage read each evening at the BAT pad, is the real number.

A REFERENCE, NOT A COPY

Huy Vector's satellite is a copper-wire cube with a clock face, a dish, a charging module beside a separate solar controller, on a brass tube over a bamboo coaster. This one keeps only the idea: a body on one post over a round base, two solar wings. Everything else is its own: a closed matte black bus instead of an open wire cube, no screen and no clock, a stainless post on a weighted brass plinth instead of a brass tube over bamboo, two panels in series into one board that is also the USB charger, a legless chip beacon on the corner where the reference has a dish, and an amber pool under the body that the reference does not have. Side by side they share a silhouette and nothing else.

HOW IT SITS WITH THE FAMILY

Black, steel and brass: one 1.5 mm stainless post like the drone's rods, a printed black body like Cyber Dawn's, a brass base like the Pocket AIs' metal, and nothing on it that is decoration. The wings are the two biggest parts and they are the power supply; the lights are a chip on the corner and the hologram's amber filament underneath, glowing on the brass. At 224 mm wide and 159 tall it stands beside the 82 mm Cyber Dawn and the 72 mm hologram as the wide one. Prototype, not product: the lid pulls off, the cell unplugs, the firmware reflashes.

Dimensions

WHAT	MM
wing tip to wing tip	224.0
each wing, along the boom	80.0
each wing, front to back	45.0
wing thickness	2.5
bus width (along the wings)	48.0
bus depth	52.0
bus height	24.0
bus wall	2.0
back lid	1.5
boom boss on each side wall	6.0
air between boss and wing	2.0
each boom, cut length	90.0

post, base to boss	110.0
post, cut length	125.0
post boss under the bus	8.0
hub on the base	8.0
hub diameter	12.0
brass base, diameter	75.0
brass base, height	8.0
cell, length (a 16340)	34.1
cell, diameter	16.6
holder, along the front wall	43.0
holder, front to back	18.0
beacon chip centre, in from the front and the right wall	5.0
pinhole for its leads, in from the right wall	6.6
beacon chip above the top face	1.1
filament under the floor, across	38.0
filament ahead of the post (negative = toward the front)	-10.0
wing plane above the desk	146.0
top of the bus above the desk	158.0
overall height to the top of the LED	159.1

Masses: wings 2 × 16 g, box and lid about 25, cell and holder 26, boards 7, rods 4.3: about 97 g above the base, 230 g of brass. Centre of mass 46 mm up, so it tips at about 39°.

Pin map, ESP32-C3 SuperMini

Five pins used of sixteen. The button keeps the family's GPIO10 and the beacon the hologram's GPIO0, the filament takes GPIO1; no strapping pin is wired, so the only resistors are the two lights'.

PIN	JOB	GOES TO
5V	power in, 3.5 to 4.0 V from the cell through the diode	SS14 cathode; anode to the charger's BAT pad
GND	ground	charger − terminal · the beacon's − lead · the filament's − lead · button
GPIO0	beacon, high = lit, PWM dims it	100 Ω, then the beacon chip's + lead, inside the box
GPIO1	downlight, high = lit, PWM dims it	68 Ω, then the filament's + lead; − lead to GND

GPIO10	mode button, active low, internal pull-up	button, other leg to GND
3V3	3.3 V out of the board's regulator	nothing (the LED takes its 3.3 V through GPIO0)
GPIO2	free: strapping pin, must float or sit high at boot	nothing
GPIO8	free: the blue LED (lit when low) and a strapping pin	nothing; firmware holds it high
GPIO9	free: the BOOT button	nothing
GPIO3 to 7, 20, 21	free	nothing (GPIO3 or 4 is the ADC pin for a cell-voltage divider, if ever wanted)
USB-C	flashing and the console only	a computer, through its own slot in the lid; safe with the cell in because of the diode

CHARGER BOARD PADS

Used: `VIN` (the wings), `G` (wing B and EN), `BAT` (to the diode), `EN` (to G), the JST-PH socket (the cell's pigtail), the USB-C port (the fallback) and the `-` screw terminal of the unused 5 V output as the controller's ground. Not used: `VU`, `D+`, `D-`, `5V`, the `+` terminal. The charge-rate jumper on the back is cut.

Every wire, and why

NET	FROM	TO	WIRE	WHY
SOL+	wing A, red lead	charger VIN pad	the panel's own 150 mm lead, along the left boom	The top of the series pair. Two 5 V panels in series read 12.2 V open-circuit in sun and still 7 to 8 V in a dim room, so they always sit above the 4.5 to 4.8 V the charger wants at its input (battery + 0.33 V, plus its own Schottky). The board takes 5 to 18 V on this pad.
SOL mid	wing A, black lead	wing B, red lead	the panels' own leads, joined inside the bus	The series link. Each wing's two leads run under its panel along the boom and through a 2.5 mm hole beside the boom socket; the link is a joint inside the box, nothing else.
GND	wing B, black lead	charger G pad	the panel's own 150 mm lead, along the right boom	The bottom of the pair. The charger's single G pad takes this wire and the EN wire; every other ground goes to the − screw terminal or the controller's GND pin.
EN	charger EN pad	charger G pad	30 AWG, 15 mm	Pulls the 5 V boost converter's enable low: the boost and its green LED stay off. The pad has a 100 kΩ pull-up to the cell rail, so this costs 37 µA, against about 1 mA if the boost and its LED ran.
VBAT	charger BAT pad	SS14 anode	30 AWG	The cell, straight: the BAT pad is the battery terminal. The load on it is 0.2 to 1.5 mA, invisible to the charger's 50 mA termination.
V5	SS14 cathode (band)	controller 5V pin	30 AWG	The controller's regulator takes 3.5 to 4.0 V here and makes 3.3 V. The diode is there for the controller's own USB-C: on this board the 5V pin is USB VBUS with no diode, so without it a flashing cable would push 5 V straight into the cell. It costs 0.15 V and nothing else.
GND	charger − screw terminal	controller GND pin	30 AWG	One ground between the two boards, in the screw terminal of the unused 5 V output so no third wire needs the G pad.

LED	GPIO0	100 Ω, then the beacon chip's + lead	the chip's own enamelled lead, tinned, plus the resistor's leg	A warm-white chip drops 2.8 to 3.0 V at a few milliamps; 100 Ω from a 3.3 V pin sets 4 to 5 mA, a clear point in a room and inside the chip's 20 mA; 68 Ω from the kit gives 7 mA if it looks faint. PWM dims it for the steady mode. GPIO0 floats briefly at boot, so the beacon may flash once; it is a lamp, so that is harmless.
GND	the beacon chip's − lead	controller GND pin	the chip's own enamelled lead, tinned	The return. The chip is glued flat on the top face at the corner; its two 0.1 mm leads run 1 mm over the surface and down a 1.2 mm pinhole, so nothing stands up but 1.1 mm of chip. The enamel burns off when the lead ends are tinned.
LAMP	GPIO1	68 Ω, then the filament's + lead	30 AWG, inside the box; the filament's own leads come up through the floor	The amber pool. A 3 V filament on a 3.3 V pin needs the resistor: about 2.5 V across the LEDs, 68 Ω, some 12 mA flat out, the hologram's figure and inside the pin's 20 mA. PWM dims it to about 3 mA for the glow mode. Both leads leave one end of the filament; they come up through a 1.5 mm hole in the floor beside it.
GND	the filament's − lead	controller GND pin	the filament's own lead, then 30 AWG	The pool's return, into the same lane as the beacon's and the button's. Never connect the filament without the resistor.
BTN	GPIO10	button leg 1	30 AWG	The mode button: off, beacon, glow, off; a long press in glow turns the pool up full. Internal pull-up, so no resistor; the family's button pin.
GND	button leg 2	controller GND pin	30 AWG	The button's return, into the same GND pin as the beacon's − lead (a 30 AWG and a hair wire fit one pin).
CELL	holder leads	JST-PH pigtail, then the charger's JST-PH socket	the holder's two leads soldered to the owned pigtail, in heat-shrink	The cell never meets an iron: the holder's leads join the pigtail 40 mm away from it. The button top is +; it goes to the pin the board marks +. Meter the plug with the cell in before it goes in.
USB	wall adapter	charger USB-C	the cable	The fallback: 500 mA into the cell, full in about 2.2 h, inside the charger's 6 h timer. The board's two Schottkys keep the panels and the cable from fighting.

The budget, honestly

From the cell: the controller in light sleep (0.13 mA), its regulator (about 0.05), the EN pull-up (0.037) and the charger (0.004) make 0.22 mA asleep. The beacon adds 4 mA for 5 % of the time blinking: 0.42 mA, 10 mAh a day, 84 days on the 950 mAh cell (850 usable) with no light at all. The glow mode adds the amber pool at 3 mA: 3.4 mA, 82 mAh a day, 10 days in the dark. The lamp mode (pool flat out at 12 mA, beacon steady) is 13.4 mA and 2.6 days: an evening setting, not a resting one. The SuperMini's red power LED, if left on, costs about 1 mA and cuts the beacon mode to 25 days.

From the wings: the charger spends 0.75 mA of input on itself while charging and gives six hours a day before its timer stops it until dark, so what the table calls stored is six hours of what the pair delivers beyond that. The figures are the listing's 90 mA taken as full sun (1 000 W/m^2; 36 cm^2 of poly cells can give little more), scaled by lux, with room LED light counted at 0.65 of daylight per lux. They are estimates until the first week on the sill is read off the BAT pad each evening.

WHERE	LUX	PAIR, MA	STORED A DAY	WHICH MEANS
desk, room light only	500	0.3	0 mAh	below what the charger itself eats
beside a window, dull day	1,000	0.9	1 mAh	barely above the charger's floor

beside a window, bright day	3,000	2.7	12 mAh	about the beacon's daily need
on the sill, overcast sky	10,000	9.0	50 mAh	beacon with margin; not the glow
an hour of direct sun	100,000	90.0	89 mAh per hour	the listing's full-sun figure; nine hours fill the cell

Parts

Three things bought on Amazon on 2026-11-03, $32.83 with tax in three orders (all three picked up by 2026-11-06; logged in `docs/INVENTORY.md`), the steel disc now spare; one brass base to buy and weight with steel; three small parts to print; everything else is in a drawer already and is listed only so it is not bought twice. Each part is named by what it is, with the link on the word to the exact product.

BOUGHT ON AMAZON 2026-11-03: THREE LINES, $32.83 WITH TAX IN THREE ORDERS

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
2 of 5	mini solar panel, 80 × 45 mm, polycrystalline, 5 V 90 mA, 150 mm leads	Amazon B0FL26KWWT (MECCANIXITY 5-pack, style 5V 0.45W 90mA)	$12.29 ($13.38 with tax); order 111-1980686-5970641 of 2026-11-03, arriving 2026-11-07	The wings, chosen by the builder on 2026-11-03 evening: a rectangle a little smaller than the reference's 110 × 60 mm panel, and Amazon's one such panel with leads. 10 poly cells under epoxy, 5 V working, about 6.1 V open circuit, 90 mA at full sun (the same as the 56 mm squares the sheet was approved with), red and black 150 mm leads welded on, so nothing is soldered on the panel. Two in series, three spare. **Incoming check:** in daylight at a window the pair reads 10 to 12 V open circuit on the meter, red positive, and each panel the same short-circuit current.
1 of 20	pre-wired warm-white LED chip, 0805	Amazon B09QGJ9LY7 (DiCUNO 20-pack, warm white)	$7.99 ($8.70 with tax); order 111-9152603-4910637 of 2026-11-03, arriving 2026-11-04	The beacon: a 2.0 × 1.25 × 1.1 mm chip that comes with two 16 cm enamelled leads already soldered on, glued flat on the top of the bus 5 mm in from the front and the right wall, its leads down a 1.2 mm pinhole beside it. 2.8 to 3.3 V, 20 mA capacity, 120°, 20 to 250 mcd. The listing says warm white and gives no colour temperature, so **check it against a 2700 K bulb before gluing**; nineteen spares to pick from. Runs at 4 to 5 mA here through 100 Ω. Tin the lead ends with a hot iron (the enamel burns off) and **meter the polarity with the diode test, which lights it faintly**, before soldering.
0 of 1	stainless disc, 3 in × 1/8 in, 304	Amazon B08JD6GY21 (Harness Machine & Fab, US made)	$9.87 ($10.75 with tax); order 111-4967394-9172244 of 2026-11-03, picked up 2026-11-06	**Not used.** The first base, 76 mm and 115 g. On the bench the steel read too white, so on 2026-11-06 the builder chose brass to match the family (next table). Spare.

TO BUY: ONE LINE

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1	brass coaster, 75 × 8 mm, flat top, hollow underneath	Amazon B0D9DMK3L7 (heavy brass coaster, 3 in, one piece)	price not read (Amazon refuses automated reads)	The base, chosen by the builder on 2026-11-06: the steel disc looked too white, and brass is the family's metal and the amber light's colour. 75 mm across and 8 mm tall, but only 80 g (2.82 oz, the listing): a thin brass top over a hollow, with a non-slip pad. Too light alone (it would tip at about 25°) and too thin to drill, so the printed hub is glued on top and about 150 g of steel is epoxied into the hollow, then the pad goes back: 230 g in all, tipping at about 39°. **Incoming check:** look underneath and measure the hollow's inside diameter and depth (they pick the steel: stacked fender washers if it is round and flat, steel BBs in epoxy if not); a magnet must not stick to the top. Clear lacquer or wax if it should stay bright.

TO PRINT: BLACK PETG, THREE SMALL PARTS (NO FILAMENT TO BUY IF A SPOOL IS ON THE SHELF)

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
1	bus box, 48 × 52 × 24 mm, 2 mm walls, open at the back	printed	about 20 g	The body, sized round the cell holder. A 6 mm boss under the floor takes the post (1.6 mm hole, 8 mm deep); a 6 mm boss on each side wall at mid-height takes a boom (1.6 mm hole, 8 mm deep through boss and wall) with a 2.5 mm wire hole beside it; a 1.2 mm pinhole in the top, 6.6 mm in from the right wall and 5 from the front, takes the beacon's leads, and the chip glues flat beside it; a 1.5 mm hole in the floor, 10 mm ahead of the post, takes the filament's leads. Inside: the holder along the front wall, the charger and the controller side by side behind it, both USB-C ports at the lid. **Check the envelope against the real holder, boards and cell before printing** (Cyber Dawn's rule); the Adafruit board's terminal block is 10 mm tall and faces the holder.
1	back lid, 44 × 20 × 1.5 mm, press fit	printed	about 3 g	The service side: two 9 × 3.4 mm slots, one for the charger's USB-C (the charge) and one for the controller's (flashing), and a 6.5 mm hole for the mode button, which is glued to the lid's inside. Neither cable needs the lid off; pull the lid to swap the cell.
1	hub, 12 mm diameter × 8 mm, 1.6 mm hole 7 mm deep	printed	about 1 g	Glued to the centre of the brass coaster's flat top with CA; the post's lower socket. The coaster's top is too thin to drill.

OWNED: ELECTRONICS

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK

1 of 2	solar charger board with 5 V boost (bq25185)	Adafruit 6106	Cyber Dawn's spare, bought 2026-10-26 (cyber_clock_v2/docs/INVENTORY.md)	**Owned.** USB-C or 5 to 18 V on the VIN pad, Schottky-ORed; 4.2 V Li-ion profile; 1 A charge, 500 mA after the jumper on the back is cut (**cut it**: Adafruit asks for 500 mA or less on this cell); holds a weak input at battery + 0.33 V and never terminates while doing so; fixed 6 h timer, reset by dark; 4 µA from the cell when idle. The 5 V boost is disabled by EN to G and its screw terminal is used only as a ground point. 29.2 × 19.1 mm, 2.5 mm holes 24.1 × 14.0 mm apart. The 50 mA display that sank Cyber Dawn's solar is not here: this load is 0.2 to 1.5 mA.
1 of 13	Nitecore 16340 cell, 950 mAh, protected, button top	Nitecore NL169; 13 owned, two IMR Batteries orders of 2026-10-16 delivered 2026-10-21	owned	**Owned.** 16.6 × 34.1 mm, 20 g, 3.42 Wh, protection under the wrap, 4.20 V termination, 2 A limit against a load of 13 mA at most. Charge at 500 mA, 0.53 C, so the jumper is cut. It lies in its holder along the front wall of the bus, button top to the right, and swaps without a tool: pull the lid, lift it out. 84 days of blinking in the dark, 10 of glow. The 500 mAh pouch the sheet was approved with goes back to Pocket AI stock.
1 of 10	16340 holder with leads	ACEIRMC, Amazon B0CRGK889F (Cyber Dawn's stock)	owned	**Owned.** A 43 × 18 × 14 mm plastic tray with two spring contacts and two leads, glued to the bus floor along the front wall. Its leads are soldered to an owned JST-PH pigtail (the 20-pair set in Pocket AI stock) in heat-shrink, 40 mm from the cell, and the plug goes into the charger's socket. **Meter the plug before it goes in: the button-top end is +.**
1 of 13	ESP32-C3 SuperMini controller	Amazon B0F888JQ91 (Meshnology 10-pack) and a 3-pack	Pocket AI stock	**Owned.** 22.5 × 18 mm, native USB, the family's board. Powered at its 5V pin from the cell through the diode. **Remove its red power LED** (the 0603 part by the USB connector, or its resistor): it would burn about 1 mA all day, five times the sleeping controller. Wi-Fi stays off. Gate it with `esptool flash_id` as before.

1 of 100	Schottky diode SS14	Chanzon, Amazon B079KJ46ZS (SMA)	drone stock, 2026-10-28	**Owned.** Inline between the charger's BAT pad and the controller's 5V pin, band toward the controller, in heat-shrink. 0.15 V at this current. It exists so the controller's USB-C can be plugged with the cell connected.
1 of 10	LED filament, 3 V, 38 mm, 2200 K	Amazon B0FR35BGYL (QIROGKC 10-pack; size 38MM, colour 2200k)	the desk hologram's pack, ordered 2026-11-01, due 2026-11-03; one used there, nine spare	**Owned.** The amber pool under the body, the warm tone that matches the family's brass: a ceramic strip of LEDs, the kind inside an Edison bulb, both leads at one end, a polarity mark. Kapton-taped flat under the box floor 10 mm ahead of the post, parallel to the wings, glowing down the post onto the brass base. GPIO1 through 68 Ω, about 12 mA flat out, PWM-dimmed. The strip is brittle: bend the leads, never the strip; never connect it without the resistor.
1 + 1	100 Ω and 68 Ω resistors, 1/4 W	LuminologyPro kit, owned · no link recorded	Pocket AI stock	**Owned.** 100 Ω in series with the beacon on GPIO0, 68 Ω with the filament on GPIO1, both inside the box in heat-shrink. 20 mA is a pin's ceiling. The two resistors in the build.
1	tactile button, 6 × 6 mm	QTEATAK kit, owned ×420 · no link recorded	Pocket AI stock	**Owned.** The mode button, glued inside the lid with its cap through the 6.5 mm hole; two 30 AWG leads to GPIO10 and GND.
about 0.4 m	30 AWG silicone wire	CBAZY kit, Amazon B073RDGTPB	Pocket AI stock	**Owned.** Seven short wires inside the box; the wings and the beacon bring their own leads. Nothing carries more than 0.5 A for an hour (the USB charge), which 30 AWG does.
	Kapton tape, white heat-shrink, CA glue, solder	ELEGOO Kapton B072Z92QZ2; the rest no links recorded	owned bench stock	**Owned.** Kapton under the boards and over the filament under the floor, heat-shrink on the diode, the two resistors and the pigtail joint, CA for the three rod sockets, the hub, the holder and the beacon chip; epoxy for the steel in the base. No flux beyond the solder's own: every joint is copper to copper, none on steel, none within 40 mm of the cell.

1	USB-C wall adapter, 2 A, and a cable	owned, no link recorded	`owned`	**Owned.** The fallback charger: into the charger board's port through the lid. Unplug when the orange light goes out; the satellite never needs it plugged in.

OWNED: FRAME

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
`3 of 10`	1.5 mm stainless rod, 150 mm coupons	uxcell 304, Amazon B082ZNMD46	`bought 2026-10-27 as coupon stock for the stainless builds`	**Owned.** One coupon cut to 125 mm is the post (7 in the hub, 110 free, 8 in the boss). The second and third give one boom each at 90 mm: 8 in the wall and yoke, 2 of air, 80 glued along the panel's back centreline with CA or epoxy. No solder anywhere in the frame.

What the firmware needs

Not written yet; listed so the pin map above is known to be enough.

- ESP-IDF 6.0.2 for the esp32c3, a new small project (no Wi-Fi, no server): `esp_pm` light sleep with the RTC timer; GPIO0 as the beacon, GPIO10 as the button with the internal pull-up, GPIO8 driven high so the blue LED stays dark.
- Three modes in NVS, cycled by a short press: off; beacon (the chip blinks 100 ms every 2 s: set the pin, sleep 100 ms, clear it, sleep 1.9 s); glow (the beacon blinks and the pool sits at about 3 mA on LEDC PWM, RC_FAST clock so it keeps running in light sleep). A long press in glow turns the pool up full and the beacon steady until the next press.
- Nothing else runs. Wi-Fi never starts; the radio's 300 mA peaks have no place in this budget. Console on USB Serial/JTAG when a cable is in.
- First-power measurements the record wants: the board's current with the power LED gone, in each mode (expected about 0.2, 0.4, 3.4 and 13 mA), the beacon's current on GPIO0 (3 to 5 mA) and the filament's on GPIO1 (about 12 mA).
- If the cell voltage is ever wanted on the beacon (a slow double-blink below 3.4 V): two resistors from BAT to GND into GPIO1, read once an hour. Not drawn; it costs two parts and 4 µA.

Before power, with the meter

- [ ] The holder's plug against the charger's socket, before it goes in: the lead from the button-top end on the pin the board marks +, the pigtail joint in heat-shrink; meter the plug with the cell in, 3.6 to 4.1 V, right way round.
- [ ] The diode's direction, before the controller is powered: band toward the controller's 5V pin; the BAT pad shows the cell's voltage, the 5V pin shows 0.15 V less. Then the panel pair at the VIN and G pads in daylight: positive, 10 to 12 V open circuit, red lead +. Then the beacon's two leads on the diode test: it lights faintly one way; mark that lead +. The filament's + mark goes toward its resistor.
- [ ] The frame floats: post to G open, each boom to G open (the wing wires run beside the booms, so a nick would show here). Everything else is a supplier spec and is not metered.

Approved, then revised for the cell

The builder approved the sheet as drawn on 2026-11-03, then chose a Nitecore 16340 cell they had bought over the 500 mAh pouch; the sheet, this record and the physics page were revised for it the same day (decision 4: the holder, the bigger bus, the two ports through the lid). Step 2, the science and engineering behind it, is published as Desk Satellite Physics: what a silicon cell does at 500 lux and why two wings go in series, how a linear charger follows a weak wing and why it eats 0.75 mA, where the 6 h timer comes from, a day on the sill in milliamp-hours, the cell, one diode and the rail, light sleep

against deep sleep, two load lines, why 2200 K looks like brass, a hundred milliseconds every two seconds, and the post as a column. All three Amazon lines were ordered that evening; the wings became 80 × 45 mm rectangles at the builder's word (decision 2), confirmed and bought the same night, arriving 2026-11-07. Next: the box is checked against the holder, the boards and the cell and printed, and the firmware is written.

Built by `docs/pages/gen/build_page.py`, which also writes `docs/FINAL_BUILD.md` and the PDFs; the dimensions, pins, nets, budget and parts live in that script. The drawing-clearance check passes. Sources read 2026-11-02: the reference page, the TI bq25185 datasheet (SLUSF65B), the Adafruit product page, Learn guide, schematic and fab print for the charger board, the Adafruit cell page, the SuperMini notes at sigmdel.ca, the Cyber Dawn and Pocket AI inventories and `cyber_clock_v2/docs/SOLAR_FEASIBILITY.md`, and the three Amazon listings. Nothing measured.